

Project paragraphs — Paul

Source outline: Project_Summery_Paul.txt

Cross-checked with: Mnist_Summery.md, PingPong_Summery.md, DND_Summery.md, CIPP_Summery.md

Purpose: One researcher-facing paragraph per internship project.

1. MNIST

The MNIST Digit Identifier served as the internship’s foundational exercise in classical machine learning and agent-assisted engineering: a canvas-based classifier that accepted a user’s handwritten digit and returned the model’s prediction, implemented with Hermes and the Gemma 4 stack on the cluster. Because the task sat squarely within a well-studied classification paradigm, the AI agent performed near-peak effectiveness when given clear context, translating high-level intent into a complete application with input and output surfaces and phased implementation steps. Technically, the work combined a trained multilayer perceptron on the MNIST catalog with a FastAPI inference service and browser UI—an end-to-end “draw → preprocess → classify” loop that also introduced the practical gap between clean training tensors and messy freehand input. More importantly for later projects, this effort solidified a research workflow in which the human supplies structured context and the agent decomposes that context into implementable phases, establishing habits that scaled to harder, less conventional problems.

2. Ping-Pong

Agent PING-PONG advanced that workflow from classical classification into learning systems grounded in existing research frameworks—mainly Gymnasium’s Atari Pong environment—to produce an AI-driven player trained against a competent computer opponent. Methodologically, the project required a disciplined multi-persona agent arrangement: an Architect (“soul”) coordinating Researcher, Coder, and Validator roles so that each iteration remained tethered to explicit objectives rather than drifting into unconstrained generation. The technical spine progressed from Behavioral Cloning on expert trajectories through Warm-Start PPO and closed-loop web play, with a parallel symbolic-training track that, in practice, yielded a learning boost of more than ten percent across training iterations relative to the earlier CNN imitation stage. In research terms, the project demonstrated both the power and the necessity of grounding agent automation in established environments and evaluation loops: elaborate plans of action were feasible precisely when personas, requirements, and training events were kept aligned, producing a fully automated training and

evaluation pipeline rather than a one-off script.

3. D&D

The Dungeons & Dragons / Hermes Protocol effort tested the limits of multi-agent collaboration under real team constraints: the mandate was a game experience driven largely by specialist AI agents, which exposed both substantial capability and clear failure modes within the available timeframe. Operationally, hallucination and drift were mitigated by stronger grounding artifacts—most notably a CHECKPOINT-style handoff discipline that kept Hermes and Gemma 4 anchored to documented truth rather than improvisation. Although the overall product was not judged a full success, the team delivered a meaningful engine substrate: centralized game-state management, a character creator, a player lobby with chat, a playable game view, and a multi-agent cast spanning monster management, rules/strategy, character identity, and the dungeon master, with the DM path capable of updating the board through the agent-and-model stack. The research lesson was unambiguous: with limited time and resources, full multilayer agent autonomy is not yet feasible for a system of this breadth, yet the same constraints clarified what agents can and cannot do alone. Moreover, deeper investment in the planning phase, plus additional calendar and compute, would likely have converted this incomplete platform into a successful demonstration.

4. CIPP (Cognitive Paradigm)

Cognitive Paradigm (CIPP) was the internship's central research objective: prior projects functioned as precursors to building an unguided, event-based learning system in which intelligence emerges from spiking dynamics rather than classical supervised or dense deep-learning defaults. Here the collaboration between student and AI agents was most strained and most instructive. Agents excelled at producing code once a path was known, but struggled when the work demanded novel research; online priors repeatedly pulled implementations back toward familiar architectures, forcing full audits or clean restarts when generated material proved unusable for a new paradigm. Mitigation came through shared plans of action and step-by-step guided workflows, yet even those controls required constant vigilance against regression to pretrained habits. The effort also pushed students into neuroscience concepts outside their prior training and revealed an infrastructure reality: stronger results often required models beyond what was self-hosted on Kant. Taken together, CIPP established both a concrete biological learning substrate and a sober finding for future work—advancing a true Cognitive paradigm demands human-led research judgment, rigorous audit, and tooling that resists the agent's gravitational pull toward what it already knows.